0580_w22_qp_43_Q6

6 **(a)** $P = 5k^2 - 7$

(i) Find the value of P when $k = 3$.

$P =$ ..... [2]

(ii) Rearrange the formula to make k the subject.

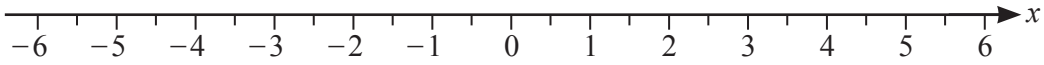
$k =$ ..... [3]

(b) **(i)** Solve.

$$x - 3 \leqslant 5x + 7$$

..... [2]

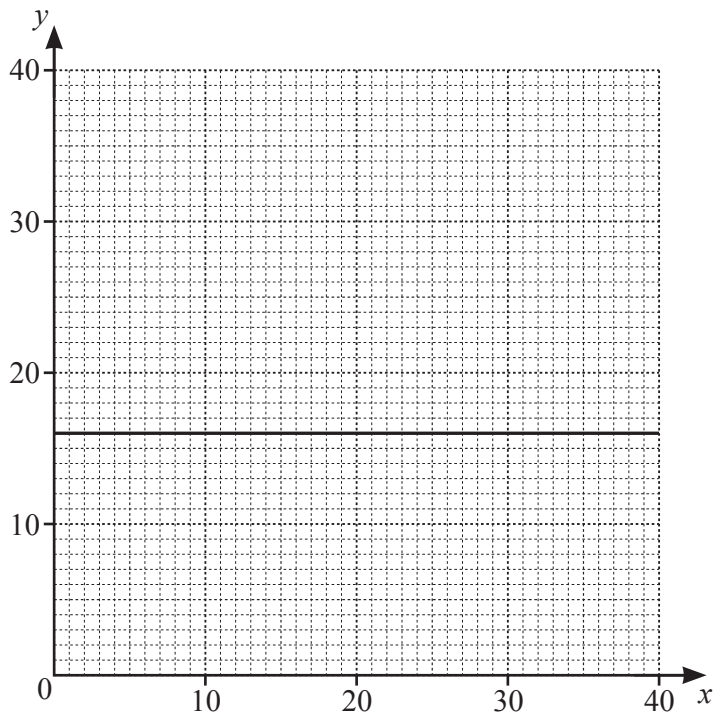
(ii) Show your answer to **part (b)(i)** on the number line.



[1]

0580_w22_qp_43_Q6

(c) The line $y = 16$ is drawn on the grid.



The region R satisfies the following inequalities.

$y \geqslant 16$ $\quad$ $x > 2$ $\quad$ $2x + 3y \geqslant 72$ $\quad$ $y \leqslant 32 - x$

(i) By drawing three more lines and shading the region **not required**, find and label region R. [6]

(ii) Find the integer coordinates (x, y) in the region R that give the maximum value of $2x + y$.

(..... , ...) [2]